

Duration: 3hrs

[Max Marks:80]

- N.B. : (1) Question No 1 is Compulsory.
(2) Attempt any three questions out of the remaining five.
(3) All questions carry equal marks.
(4) Assume suitable data, if required and state it clearly.

- 1 Attempt any FOUR [20]
a What is low level and high level modulation?
b Define: a) Signal to Noise Ratio b) Selectivity c) Sensitivity
c Why is VSB amplitude modulation used in television broadcasting?
d What is aperture effect? How to avoid it?
e What is multiplexing? State its advantages.
- 2 a How is FET reactance modulator capable of generating FM signal? Use neat circuit diagram to explain the same. [10]
b Derive the wave equation for AM wave. Draw the time domain and frequency domain representation of AM wave. [10]
- 3 a Explain the working of ratio detector as FM demodulator. What are its advantages over balanced slope detector? [10]
b What is sampling? State and prove sampling theorem for low pass signals. [10]
- 4 a Explain TDM transmitter and receiver block diagram. [10]
b Explain the working of ISB receiver. [10]
- 5 a Define noise factor and noise figure. Determine the overall noise factor and noise figure for three cascaded amplifiers with the following parameters: $A_1 = 3\text{dB}$, $A_2 = 13\text{dB}$, $A_3 = 10\text{dB}$, $NF_1 = 10\text{dB}$, $NF_2 = 6\text{dB}$, $NF_3 = 10\text{dB}$ [10]
b Explain the working of diode detector. How is practical diode detector different from diode detector? [10]
- 6 a Explain indirect method of FM generation with the help of relevant phasor diagrams [10]
b Explain DPCM [5]
c Compare PAM, PWM and PPM [5]

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- 1 Attempt any **FOUR** [20]
a Explain the necessity of de-emphasis and pre-emphasis
b Compare AM and FM.
c What is aliasing? How it can be avoided?
d A transmitter radiates 9kW of power with carrier unmodulated and 10.125 kW when modulated. Calculate the depth of modulation.
e List the advantages of pulse modulation over continuous modulation schemes.
- 2 a Explain indirect FM transmitter. [10]
b Draw a neat block diagram of a superheterodyne radio receiver and explain each block in detail. [10]
- 3 a What are the different methods for SSB generation? Explain any one in detail. [10]
b Explain the balanced slope detector with the help of a schematic diagram. [10]
- 4 a State and prove sampling theorem for low pass bandlimited signal. [10]
b With the help of suitable waveforms explain the generation and detection of PPM. [10]
- 5 a Draw and explain the FDM transmitter & receiver block diagram along with its applications. [10]
b With the help of a block diagram explain the concept of PCM. [10]
- 6 a Explain in detail the balanced modulator with suitable expressions and waveforms. [10]
b i) Write a note on Delta and adaptive delta modulation
ii) Explain VSB in broadcast television. [10]

University of Mumbai

Examination Summer 2022

Program: Electronics and Telecommunication Engineering

Curriculum Scheme: Rev2019

Examination: **SE Semester: IV**

Course Code: **ECC405** and Course Name: **Principles of Communication Engineering**

Time: 2 hour 30 minutes DATE: 30/5/2022

QP CODE:94013 Max. Marks: 80

Q1.	Choose the correct option for the following questions. All the questions are compulsory and carry equal marks
1.	Medium which sends information from source to receiver is called
Option A:	Channel
Option B:	Transmitter
Option C:	Loudspeaker
Option D:	Transducer
2.	What is modulation?
Option A:	Process of separating a carrier signal and analog signal
Option B:	Recovering information from a modulated signal
Option C:	Process of varying one or more properties of a modulating signal
Option D:	Involvement of noise
3.	The ability of receivers to select the wanted signals among various incoming signals is called
Option A:	Fidelity
Option B:	Selectivity
Option C:	Sensitivity
Option D:	Modulation
4.	For low level amplitude modulation, amplifier must be
Option A:	Class C amplifier
Option B:	Class B amplifier
Option C:	Class D amplifier
Option D:	Class A amplifier
5.	In a radio receiver, noise is generally developed at
Option A:	IF stage
Option B:	Receiving antenna
Option C:	Audio stage
Option D:	RF stage
6.	A superheterodyne receiver with an IF of 450 kHz is tuned to a signal at 1250 kHz. The image frequency is
Option A:	1700 kHz
Option B:	2150 kHz
Option C:	1650 kHz
Option D:	2100 kHz

7.	Standard intermediate frequency used for AM receiver is
Option A:	455 MHz
Option B:	455 KHz
Option C:	455 Hz
Option D:	20 KHz
8.	Pre-Emphasis Circuit is used to amplify what kind of frequencies?
Option A:	Low
Option B:	High
Option C:	Moderate
Option D:	Oscillator
9.	Natural and flat top sampling are the types of
Option A:	PWM
Option B:	PPM
Option C:	PCM
Option D:	PAM
10.	Which is a circuit used to generate a double sideband suppressed carrier signal?
Option A:	Sideband suppressor
Option B:	Anti-modulator
Option C:	Balanced modulator
Option D:	Carrier suppressor

Q2.	Solve any four Questions out of Six, (5 marks each)
A	Explain the use of VSB in broadcast television
B	A transmitter radiates 9kW of power with carrier unmodulated and 10.125 kW when modulated. Calculate the depth of modulation.
C	Explain types of AGC.
D	Define and explain SNR, Noise Figure, Noise factor, Noise Temperature, and Friss Formula.
E	Explain how PPM is generated from PWM?
F	Explain FDM and TDM applications

Q3.	Solve any Two Questions out of Three (10 marks each)
A	Draw a neat block diagram of a superheterodyne radio receiver and explain each block in detail.
B	What are different methods for SSB generation? Explain any one method in detail.
C	With the help of a block diagram explain the concept of PCM. What is DPCM?

Q4.	Solve any Two Questions out of Three (10 marks each)
A	What are different methods for FM demodulation? Explain any one method in detail.
B	State and prove sampling theorem for low pass band-limited signal.
C	Explain Frequency Division Multiplexing (FDM) transmitter & receiver with a neat block diagram.